

Session 2 summary

Source links embedded in the EPUB

UNSAFE_HTML_20260622181236.html: https://blackboard.ie.edu/bbcswebdav/pid-2761814-dt-content-rid-66131270_1/xid-66131270_1?Kq3cZcYS15=6671a827ba5c4c2183a4e13eef765cf0&VxJw3wfC56=1782578738&3cCnGYSz89=hpN5y4Q2jZxhoaKtUEZhBkjP4iyUY%2Fb5FggQQ45db58%3D

Summary Session 2



AI Challenge · Session 2 Summary

AI foundations: models, memory, tokenomics, hallucinations, and image generation



WHY THIS SESSION MATTERS

- Students moved from using AI casually to understanding how generative AI actually works.
- The session focused on choosing the right model for the right business task.
- A key goal was learning why memory, token costs, and hallucinations matter in companies.



MEET THE SESSION FOCUS



LLMs & SLMs



Generative AI



Memory & tokens



Hallucinations



Image generation



KEY IDEAS DISCUSSED

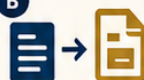
A



LLMs vs SLMs

Frontier LLMs push the benchmarks but are expensive. SLMs are smaller, cheaper, faster, and often better for narrow company tasks such as classification.

B



Generative vs traditional AI

Traditional AI transforms or classifies data. Generative AI creates new text, code, images, audio, or video.

C



How GenAI works

The class used GANs to explain the core idea: a generator creates, a discriminator critiques, and AI behaves like a probability machine.

D



Transformers & memory

Transformers help AI keep track of context in a conversation. Memory is limited and must fit inside the context window.

E



Tokenomics matters

Every prompt and answer consumes tokens. More context means more cost, so companies must optimize memory and prompt design.

F



Hallucinations & black box

AI can sound convincing while being wrong. Since the model is a black box, human oversight and strong context remain essential.



WHEN TO USE WHICH AI

Use generative AI for probabilistic tasks

- ✓ Ideation
- ✓ Drafting text
- ✓ Creative design
- ✓ Image creation
- ✓ Flexible classification with variation

VS

Prefer traditional or deterministic systems for precise tasks

- ✓ Calculations
- ✓ Transactions
- ✓ Fixed-rule processes
- ✓ High-precision operations
- ✓ Anything that must be consistently exact



Rule of thumb: if a deterministic tool can solve it reliably, use that first.



IMAGE GENERATION EXPLAINED

1

Mask / missing pixels



2

AI guesses what should be there



3

Diffusion model creates a new image



The session introduced diffusion models and showed how modern tools generate realistic images from prompts or partial visual clues.



LIVE ACTIVITIES

1

ChatGPT memory self-portrait

Students asked ChatGPT to visualize what it knew about them.



2

AI or human?

Students played a recognition game to guess whether images were AI-generated or human-made.



3

Create an image / figurine

Students experimented with image generation tools such as ChatGPT or Gemini.



IMPORTANT REMINDERS



Participation depends on submitting the in-class activities in Blackboard.



Some Blackboard tasks were individual because groups were not yet configured.



Wrong memory can pollute future prompts, so reviewing AI memory is useful.



Human supervision is necessary to detect and correct hallucinations.



TAKEAWAYS



AI is a probability machine, not a human mind.



Smaller models are often the smarter business choice.



Memory and context improve results but increase cost.



Generative AI shines in creative work, not deterministic accuracy.



AI Challenge · Session 2 · Blackboard recap